Further modification of a generalised 3D Hoek–Brown criterion: the GZZ criterion

H. CHEN*, H. ZHU† and L. ZHANG‡

This paper presents a further modification of the generalised three-dimensional (3D) Hoek–Brown (HB) criterion – the generalised 3D Zhang-Zhu (GZZ) criterion. Specifically, a modification was made by introducing a 3D version of effective mean stress to replace the original two-dimensional (2D) version of effective mean stress in the GZZ criterion. The modification overcomes the non-smoothness and non-convexity problems of the GZZ criterion and maintains a simple form of the criterion expression. The modified GZZ criterion can also reduce to the 2D HB criterion under both triaxial compression and extension conditions and thus inherits the advantages of and uses the same parameters as the HB criterion. Eleven sets of true triaxial test data were collected from publications to evaluate the newly modified GZZ criterion. The results indicate that the newly modified GZZ criterion has the highest or at least the same accuracy for strength prediction among the various 3D versions of the HB criterion that can reduce to the HB criterion and are smooth and convex. Moreover, of the various 3D versions of the HB criterion that can reduce to the HB criterion and are smooth and convex, only the newly modified GZZ criterion can be expressed explicitly.

KEYWORDS: failure; rocks/rock mechanics; shear strength

NOTATION

D	disturbance factor
$I_1,\ I_2,\ I_3$	first, second and third stress invariants
$I_1^*,\ I_2^*,\ I_3^*$	transformed first, second and third stress invariants
$J_1,\ J_2,\ J_3$	first, second and third deviatoric stress invariants
$J_1^*,\ J_2^*,\ J_3^*$	transformed first, second and third deviatoric stress invariants
$k,\ \delta$	aspect ratio of octahedral stress under triaxial compression and extension status
m_b	constant for the rock mass in the Hoek–Brown (HB) criterion
m_i	constant for the intact rock in HB criterion
p^*	$1/3I_1^*$
q	$\sqrt{I_1^{*2}-3I_2^*}$
$s,\ a$	constants for rock reflecting fracture and rock mass conditions
η	q/p^*
$\theta_\sigma,\ \theta$	Lode's angle measured from x- and y-axis
ρ	radius of criterion in π-plane
$\sigma_1',\ \sigma_2',\ \sigma_3'$	major, intermediate and minor effective principal stresses
$\sigma_1^*,\ \sigma_2^*,\ \sigma_3^*$	transformed major, intermediate and minor effective principal stresses
σ_c	unconfined compressive strength of the intact rock
σ_m'	three-dimensional version of $\sigma_{\mathrm{m},2}{}'$
$\sigma_{\mathrm{m},2}'$	effective mean stress
$\sigma_\mathrm{n}',\ \tau$	effective normal stress and shear stress on failure surface
$\sigma_x',\ \sigma_y',\sigma_z',\tau_{xy},\tau_{yz},\tau_{zx}$	six effective stress components in Cartesian coordinates
τ_oct	octahedral shear stress
$\phi_\mathrm{t},\ c_\mathrm{t}$	tangential friction angle and cohesion on failure surface

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*Department of Civil and Architectural Engineering and Mechanics, University of Arizona, Tucson, AZ, USA (Orcid:0000-0003-3482-2283).
†Department of Geotechnical Engineering, Tongji University, Shanghai, China.
‡Department of Civil and Architectural Engineering and Mechanics, University of Arizona, Tucson, AZ, USA (Orcid:0000-0003-0969-1226).

INTRODUCTION

The Hoek–Brown (HB) strength criterion is the most widely used empirical criterion for intact rocks and jointed rock masses (Hoek & Brown, 1980, 1988, 1997; Hoek *et al.*, 2002). The original HB criterion can be expressed as (Hoek & Brown, 1980)

$$\sigma_1' = \sigma_3' + \sigma_\mathrm{c}\sqrt{m_\mathrm{i}\frac{\sigma_3'}{\sigma_\mathrm{c}}+1} \tag{1}$$

where σ_1' and σ_3' are major and minor effective principal stresses; σ_c is the unconfined compressive strength of intact rock; and m_i is a constant for rock.

To take the effects of fracture and rock mass conditions into account, the original HB criterion was revised to a generalised form as (Hoek, 1994)

$$\sigma_1' = \sigma_3' + \sigma_\mathrm{c}\left(m_\mathrm{b}\frac{\sigma_3'}{\sigma_\mathrm{c}}+s\right)^a \tag{2}$$

where m_b, a, s are constants reflecting fracture and rock mass conditions, and are estimated based on the geological strength index (GSI) (Hoek *et al.*, 2002)

$$m_\mathrm{b} = \exp\left(\frac{\mathrm{GSI}-100}{28-14D}\right)m_\mathrm{i} \tag{3a}$$

$$s = \exp\left(\frac{\mathrm{GSI}-100}{9-3D}\right) \tag{3b}$$

$$a = 0{\cdot}5 + \frac{1}{6}\left(e^{-\mathrm{GSI}/15} - e^{-20/3}\right) \tag{3c}$$

where D is a factor representing the degree of disturbance to rock masses caused by blast damage and stress relaxation.

Although the HB criterion has been widely used, it does not consider the intermediate principal stress σ_2' in either its original or generalised form. To consider σ_2', different methods have been proposed for extending the HB criterion to a three-dimensional (3D) form (Pan & Hudson, 1988; Priest, 2005; Zhang & Zhu, 2007; Jiang *et al.*, 2011). Among those 3D criteria, the generalised 3D HB criterion proposed by Zhang & Zhu (2007) and Zhang (2008), called the GZZ criterion from now on, can successfully reduce to the HB criterion under both triaxial compression (TC) and triaxial extension (TE) conditions and produce a good fit to true triaxial test data of a wide range of rock types (Bertuzzi *et al.*, 2017). The GZZ criterion is expressed as

$$\frac{1}{\sigma_{\mathrm{c}}^{(1/a-1)}}\left(\frac{3}{\sqrt{2}}\tau_{\mathrm{oct}}\right)^{1/a} + \frac{m_{\mathrm{b}}}{2}\left(\frac{3}{\sqrt{2}}\tau_{\mathrm{oct}}\right) - m_{\mathrm{b}}\sigma_{\mathrm{m},2}' = s\sigma_{\mathrm{c}} \tag{4}$$

where

$$\tau_{\mathrm{oct}} = \frac{1}{3}\sqrt{(\sigma_1' - \sigma_2')^2 + (\sigma_2' - \sigma_3')^2 + (\sigma_3' - \sigma_1')^2} \tag{5a}$$

$$\sigma_{\mathrm{m},2}' = \frac{\sigma_1' + \sigma_3'}{2} \tag{5b}$$

The GZZ criterion uses the same parameters as the HB criterion and hence the methods for determining the HB criterion parameters can be simply adopted when using the GZZ criterion. However, the GZZ criterion is concave at the TE state and not smooth at either TC or TE, which may result in problems during numerical analysis.

To tackle the non-smoothness and non-convexity problems of the GZZ criterion, Zhang *et al.* (2013) adopted three smooth and convex Lode dependences to the GZZ criterion without losing the accuracy on strength prediction. However, using the Lode dependences is complicated because no explicit expression is available, and a sophisticated iterative algorithm is needed for the case when $a \neq 0{\cdot}5$.

Therefore, in this paper, the GZZ criterion is further modified by using a method that is much simpler than that used by Zhang *et al.* (2013). Specifically, a 3D version of $\sigma_{\mathrm{m},2}'$ is introduced to replace the original 2D $\sigma_{\mathrm{m},2}'$ in the GZZ criterion. The modification successfully converts the GZZ criterion to a simple equation using three (translated) stress invariants, which efficiently solves the non-smoothness and non-convexity problems. Moreover, the simple form of the newly modified GZZ criterion avoids the use of the Lode-dependence function and iterative algorithm. Eleven sets of true triaxial test data were collected from the literature to verify the newly modified GZZ criterion.

BRIEF REVIEW OF 3D VERSIONS OF HB CRITERION

Many authors proposed 3D versions of the HB criterion to consider σ_2'. Table 1 lists some of the important ones. These 3D HB criteria can be classified into three categories as shown in Fig. 1: criteria by incorporating a Lode function (Jiang *et al.*, 2011; Jiang & Xie, 2012; Jiang & Zhao, 2015; Jiang, 2017a, 2017b), criteria by adopting Lode dependences (Benz *et al.*, 2008; Lee *et al.*, 2012, Zhang *et al.*, 2013; Wu *et al.*, 2018) and criteria by introducing $\sigma_{\mathrm{m},2}'$ (Zhang & Zhu, 2007; Zhang, 2008).

Table 2 briefly reviews the various 3D HB criteria, focusing on smoothness, convexity and reduction to the HB criterion. Note that none of the 3D HB criteria can extend the HB criterion to a true 3D version with a smooth and convex shape and an explicit expression. Among them, the GZZ criterion has a simple and explicit formulation and can reduce to the HB criterion but is neither smooth nor convex at the TE condition. Therefore, this paper will further modify the GZZ criterion to make it smooth and convex while keeping the current expected features.

FURTHER MODIFICATION OF GZZ CRITERION

In this part, a simple method is introduced to address the non-smoothness and non-convexity problems of the GZZ criterion. The other consideration is that the newly modified GZZ criterion can still be expressed explicitly and reduce to the HB criterion at both TC and TE conditions.

The GZZ criterion successfully extends the HB criterion to a 3D version by introducing τ_{oct} and $\sigma_{\mathrm{m},2}'$. Utilisation of the $\sigma_{\mathrm{m},2}'$, is the main reason for the non-smoothness and non-convexity problems at TE/TC conditions. The direction of σ_1' and σ_3' changes abruptly when the stress passes TE/TC and thus $\sigma_{\mathrm{m},2}'$ is not invariant with respect to σ_x', $\sigma_y', \sigma_z', \tau_{xy}, \tau_{yz}$ and τ_{zx}. This leads to the non-smoothness of the GZZ criterion at TE/TC. Hence, finding a unified modification of $\sigma_{\mathrm{m},2}'$ is the key to smoothening the GZZ criterion. The other problem is the non-convexity of the GZZ criterion under TE as shown in Fig. 1. According to Jiang & Pietruszczak (1988), the convexity of a function $\rho = \rho(\theta)$ is preserved if the following condition is satisfied

$$g = \rho^2 + 2\left(\frac{\partial \rho}{\partial \theta}\right)^2 - \rho\frac{\partial^2 \rho}{\partial \theta^2} \geq 0 \tag{6}$$

where $\rho(\theta)$ is the radius function of the criterion in π-plane. The convexity of the proposed criterion will be further discussed in section 'Smoothness and convexity'.

In solid mechanics, the three stress invariants, I_1, I_2 and I_3, remain the same regardless of the stress conditions and are often used in 3D strength criteria such as the SMP criterion (Matsuoka, 1974) and the Lade criterion (Lade & Duncan, 1975). Therefore, the stress invariants as defined below are adopted to further modify the GZZ criterion

$$I_1 = \sigma_1' + \sigma_2' + \sigma_3' = \sigma_{kk}' = \mathrm{tr}(\sigma); k = x, y, z \tag{7a}$$

$$I_2 = \sigma_1'\sigma_2' + \sigma_2'\sigma_3' + \sigma_3'\sigma_1' = \frac{1}{2}\left(\sigma_{ii}'\sigma_{jj}' - \sigma_{ij}'\sigma_{ji}'\right); \quad i, j = x, y, z \tag{7b}$$

$$I_3 = \det(\sigma) = \det\left(\sigma_{ij}'\right); i, j = x, y, z \tag{7c}$$

$$\boldsymbol{\sigma} = \left[\sigma_{ij}'\right] = \begin{bmatrix} \sigma_{xx}' & \sigma_{xy}' & \sigma_{xz}' \\ \sigma_{yx}' & \sigma_{yy}' & \sigma_{yz}' \\ \sigma_{zx}' & \sigma_{zy}' & \sigma_{zz}' \end{bmatrix} = \begin{bmatrix} \sigma_x' & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_y' & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_z' \end{bmatrix} \tag{7d}$$

Under TE and TC conditions, the three invariants can be written as equations (8) and (9), respectively

$$I_1 = \sigma_1' + \sigma_2' + \sigma_3' = \sigma_1' + 2\sigma_3' \tag{8a}$$

$$I_2 = \sigma_1'\sigma_2' + \sigma_2'\sigma_3' + \sigma_3'\sigma_1' = (2\sigma_1' + \sigma_3')\sigma_3' \tag{8b}$$

$$I_3 = \sigma_1'\sigma_2'\sigma_3' = \sigma_1'\sigma_3'^2 \tag{8c}$$

Table 1. 3D versions of HB criterion

Source	Expression
Pan–Hudson criterion (Pan & Hudson, 1988)*	$\frac{3J_2}{\sigma_c^2}+\frac{\sqrt{3}m_i}{2}\frac{\sqrt{J_2}}{\sigma_c}-\frac{m_i}{3}\frac{I_1}{\sigma_c}=s$
Priest criterion (Priest, 2005, 2012)	$J_2^{1/2}=AI_1+B;\ A=\frac{3\sqrt{J_2}-\sigma_{cm}\sqrt{3}}{3J_1-\sigma_{cm}},\ B=\frac{\sigma_{cm}\left(\sqrt{3}-A\right)}{3}$
Jiang criteria (Jiang *et al.*, 2011; Jiang & Xie, 2012; Jiang & Zhao, 2015; Jiang, 2017a, 2017b)†	$\frac{3J_2}{\sigma_{ci}^2}+A(\theta)\frac{m_i}{\sqrt{3}}\frac{\sqrt{J_2}}{\sigma_{ci}}-\frac{m_i}{3}\frac{I_1}{\sigma_{ci}}=s$
Criterion by Benz *et al.* (2008)‡	$f_{HB,MN}=q-L\ M_{c,HB}(p'-p^*);\ p'=\frac{\sigma_1'+\sigma_2'+\sigma_3'}{3},\ q=\sqrt{3J_2}$
Criterion by Lee *et al.* (2012)§	$F_{HB\text{-}WW}=\rho-g_{ww}(\theta,\xi)\rho_c;\ \xi=\frac{\sigma_1+\sigma_2+\sigma_3}{\sqrt{3}},\ \rho=\sqrt{2J_2}$
Modified GZZ criterion (Zhang *et al.*, 2013)‖	$\sqrt{J_2}=L(\theta_\sigma)_{X-D}\sqrt{J_{2,\max}}$

*$I_1=\sigma_1'+\sigma_2'+\sigma_3'$ and $J_2=\frac{1}{6}\left[(\sigma_1'-\sigma_2')^2+(\sigma_2'-\sigma_3')^2+(\sigma_3'-\sigma_1')^2\right]=\frac{3}{2}\tau_{oct}^2$.

†$A(\theta)$ is an empirical Lode angle-dependent function.

$$‡L(\theta_\sigma)=\frac{\sqrt{3}\delta}{2\sqrt{\delta^2-\delta+1}}\frac{1}{\cos\psi}\begin{cases}\psi=\frac{1}{6}\arccos\left[-1+\frac{27\delta^2(1-\delta)^2}{2(\delta^2-\delta+1)^3}\sin^2(3\theta_\sigma)\right], & \text{for } \theta_\sigma\geq 0\\ \psi=\frac{\pi}{3}-\frac{1}{6}\arccos\left[-1+\frac{27\delta^2(1-\delta)^2}{2(\delta^2-\delta+1)^3}\sin^2(3\theta_\sigma)\right], & \text{for } \theta_\sigma<0\end{cases};$$

$$\delta=\frac{M_{e,HB}}{M_{c,HB}};\ \theta_\sigma=\frac{1}{3}\sin^{-1}\left(-\frac{3\sqrt{3}J_3}{2J_2^{3/2}}\right);\ p^*=\begin{cases}p_t^s=-\frac{\sigma_{ci}s}{m_b}, & \text{for } M_{c.HB}=M_{c,HB}^s\\ p_t^t=-\frac{\tilde{f}}{M_{c,HB}}-\frac{\tilde{f}}{3}+\sigma_3, & \text{for } M_{c.HB}=M_{c,HB}^t\end{cases}.$$

$$§g_{ww}(\theta,\xi)=\frac{-(\pi/3)+(2k-1)\sqrt{4(1-k^2)\cos^2(\theta-\pi/3)+k(5k-4)}}{4(1-k^2)\cos^2(\theta-\pi/3)+(2k-1)^2};\ k=g_{ww}\left(\frac{\pi}{3}\right)=\frac{\rho|_{\theta=\pi/3}}{\rho|_{\theta=0}};\ \theta=\frac{1}{3}\cos^{-1}\left(\frac{3\sqrt{3}J_3}{2J_2^{3/2}}\right).$$

‖Three Lode dependences with smoothness and convexity characteristics, elliptical dependence (E-D) based on William & Warnke (1975), hyperbolic dependence (H-D) based on Yu *et al.* (2002), and spatial mobilised plane dependence (S-D) based on Matsuoka (1974) were adopted:

$$L(\theta_\sigma)_{E\text{-}D}=\frac{2(1-\delta^2)\cos(\pi/6-\theta_\sigma)+(2\delta-1)\sqrt{4(1-\delta^2)\cos^2(\pi/6-\theta_\sigma)+\delta(5\delta-4)}}{4(1-\delta^2)\cos^2(\pi/6-\theta_\sigma)+(2\delta-1)^2};\ \delta=\sqrt{J_{2,\min}/J_{2,\max}}=\sqrt{J_2(-\pi/6)/J_2(\pi/6)};$$

$$L(\theta_\sigma)_{H\text{-}D}=\frac{2\delta(1-\delta^2)\cos(\pi/6+\theta_\sigma)+\delta(\delta-2)\sqrt{4(\delta^2-1)\cos^2(\pi/6+\theta_\sigma)+(5-4\delta)}}{4(1-\delta^2)\cos^2(\pi/6+\theta_\sigma)-(\delta-2)^2};$$

$$L(\theta_\sigma)_{S\text{-}D}=\frac{\sqrt{3}\delta}{2\sqrt{\delta^2-\delta+1}}\frac{1}{\cos\psi}\begin{cases}\psi=\frac{1}{6}\arccos\left[-1+\frac{27\delta^2(1-\delta)^2}{2(\delta^2-\delta+1)^3}\sin^2(3\theta_\sigma)\right], & \text{for } \theta_\sigma\geq 0\\ \psi=\frac{\pi}{3}-\frac{1}{6}\arccos\left[-1+\frac{27\delta^2(1-\delta)^2}{2(\delta^2-\delta+1)^3}\sin^2(3\theta_\sigma)\right], & \text{for } \theta_\sigma<0\end{cases}.$$

$$I_1=\sigma_1'+\sigma_2'+\sigma_3'=2\sigma_1'+\sigma_3' \quad (9a)$$

$$I_2=\sigma_1'\sigma_2'+\sigma_2'\sigma_3'+\sigma_3'\sigma_1'=(\sigma_1'+2\sigma_3')\sigma_1' \quad (9b)$$

$$I_3=\sigma_1'\sigma_2'\sigma_3'=\sigma_1'^2\sigma_3' \quad (9c)$$

For both cases, the second deviatoric stress invariant J_2 and I_1I_2/I_3 can be simplified as

$$J_2=\frac{1}{6}\left[(\sigma_1'-\sigma_2')^2+(\sigma_2'-\sigma_3')^2+(\sigma_3'-\sigma_1')^2\right]=\frac{1}{3}(\sigma_1'-\sigma_3') \quad (10a)$$

$$\frac{I_1I_2}{I_3}=\frac{(\sigma_1'+\sigma_2'+\sigma_3')(\sigma_1'\sigma_2'+\sigma_2'\sigma_3'+\sigma_3'\sigma_1')}{\sigma_1'\sigma_2'\sigma_3'}=\frac{(2\sigma_1'+\sigma_3')(\sigma_1'+2\sigma_3')}{\sigma_1'\sigma_3'}=1+\frac{(\sigma_1'+\sigma_3')^2}{\sigma_1'\sigma_3'} \quad (10b)$$

Furthermore, $\sigma_{m,2}'$ can be rewritten as

$$\sigma_{m,2}'=\frac{\sigma_1'+\sigma_3'}{2}=\sqrt{\left(\frac{\sigma_1'-\sigma_3'}{2}\right)^2+\sigma_1'\sigma_3'}=\sqrt{\frac{3}{4}J_2+\sigma_1'\sigma_3'} \quad (11)$$

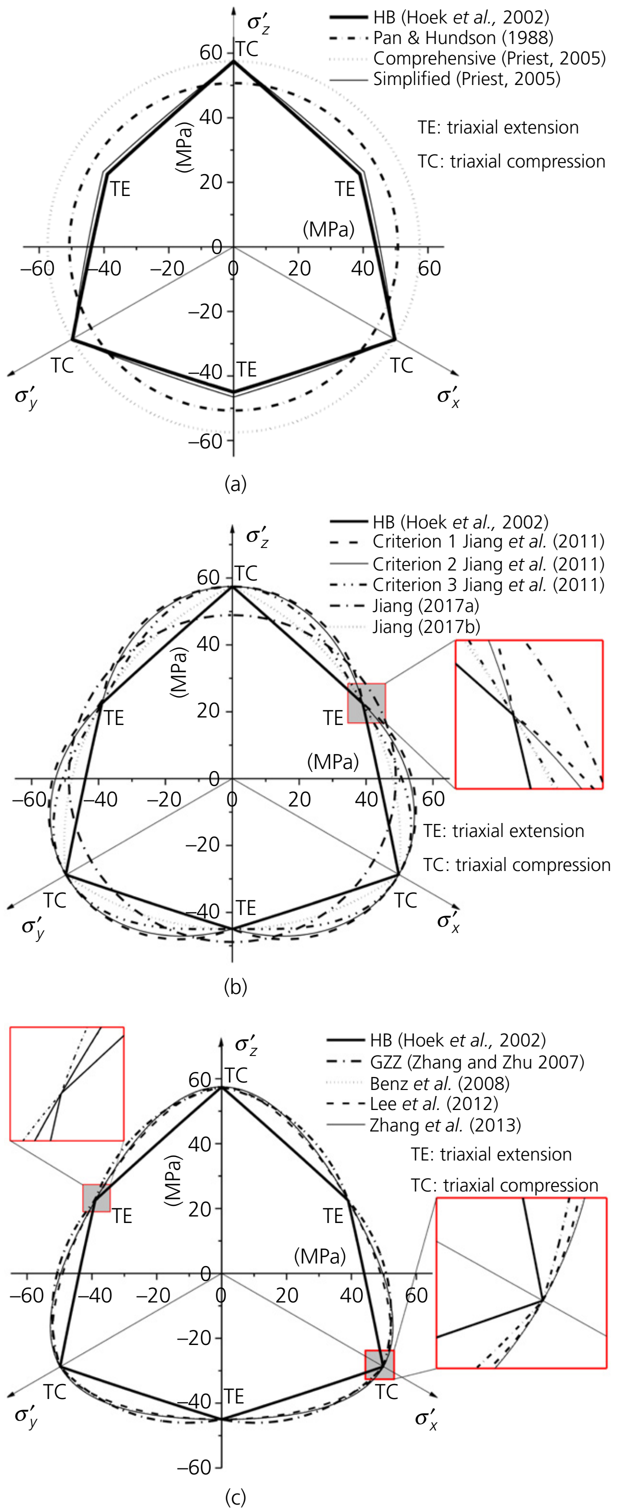

Fig. 1. Comparison between HB criterion and 3D version of HB criterion by (a) Pan & Hudson (1988) and Priest (2005); (b) Jiang *et al.* (2011) and Jiang (2017a, 2017b) and (c) Zhang & Zhu (2007), Benz *et al.* (2008), Lee *et al.* (2012) and Zhang *et al.* (2013) at $I_1 = 100$ MPa for a rock with $\sigma_c = 50$ MPa, $m_i = 5$, $s = 0{\cdot}01$ and $a = 0{\cdot}5$ in π plane

Combining equations (10) and (11), $\sigma_1'\sigma_3'$ can be rewritten as

$$\sigma_1'\sigma_3' = \frac{18J_2I_3}{I_1^3 - 3I_1J_2 - 27I_3} = \frac{2\left(I_1^2I_3 - 3I_2I_3\right)}{I_1I_2 - 9I_3} \tag{12}$$

and equation (11) can be rewritten as

$$\begin{aligned}\sigma_m' &= \frac{\sigma_1' + \sigma_3'}{2} = \frac{1}{2}\sqrt{\frac{3I_1^3J_2 - 9I_1J_2^2 - 9I_3J_2}{I_1^3 - 3I_1J_2 - 27I_3}} \\ &= \frac{1}{2}\sqrt{\frac{18I_1^3\tau_{oct}^2 - 81I_1\tau_{oct}^4 - 54I_3\tau_{oct}^2}{4I_1^3 - 18I_1\tau_{oct}^2 - 108I_3}}\end{aligned} \tag{13}$$

The σ_m' in equation (13) reduces to $\sigma_{m,2}'$ under both TE and TC and thus can be seen as a general expression of $\sigma_{m,2}'$. By using σ_m', equation (4) can be modified as

$$\begin{aligned}&\frac{1}{\sigma_c^{(1/a-1)}}\left(\frac{3}{\sqrt{2}}\tau_{oct}\right)^{1/a} + \frac{m_b}{2}\left(\frac{3}{\sqrt{2}}\tau_{oct}\right) \\ &- \frac{m_b}{2}\sqrt{\frac{18I_1^3\tau_{oct}^2 - 81I_1\tau_{oct}^4 - 54I_3\tau_{oct}^2}{4I_1^3 - 18I_1\tau_{oct}^2 - 108I_3}} = s\sigma_c\end{aligned} \tag{14}$$

Under unconfined compression (UC) condition, the denominator of σ_m' becomes zero and σ_m' will be meaningless. Therefore, an isotropic tensile strength $t_0 = (\sigma_c/m_b)s$ is introduced to convert the proposed criterion. As shown in Fig. 2, σ_3' and σ_2' would no longer be zero under the UC condition if the $p' - q$ coordinates are translated along the p' axis leftward by t_0.

By defining the translated stress as the summation of the effective stress and t_0 as

$$\sigma_i^* = \sigma_i' + \frac{\sigma_c}{m_b}s;\ i = 1, 2, 3 \tag{15}$$

the newly modified GZZ criterion can be rewritten as

$$\begin{aligned}&\frac{1}{\sigma_c^{(1/a-1)}}\left(\frac{3}{\sqrt{2}}\tau_{oct}\right)^{1/a} + \frac{m_b}{2}\left(\frac{3}{\sqrt{2}}\tau_{oct}\right) \\ &- \frac{m_b}{2}\sqrt{\frac{18I_1^{*3}\tau_{oct}^2 - 81I_1^*\tau_{oct}^4 - 54I_3^*\tau_{oct}^2}{4I_1^{*3} - 18I_1^*\tau_{oct}^2 - 108I_3^*}} = 0\end{aligned} \tag{16}$$

where

$$I_1^* = \sigma_1^* + \sigma_2^* + \sigma_3^* = I_1 + 3\frac{\sigma_c}{m_b}s \tag{17a}$$

$$I_3^* = \sigma_1^*\sigma_2^*\sigma_3^* \tag{17b}$$

EVALUATION OF NEWLY MODIFIED GZZ CRITERION

Smoothness and convexity

In π-plane, the newly modified GZZ criterion is a curved triangle and can be expressed by Lode angle. Particularly, the Lode angle can be expressed by stress invariants as

$$\cos(3\theta) = \frac{3\sqrt{3}}{2}\frac{J_3^*}{J_2^{3/2}} = \frac{3\sqrt{3}}{2}\frac{J_3^*}{J_2^{3/2}} \tag{18a}$$

$$J_3^* = \frac{27I_3^* + 9I_1^*J_2 - I_1^{*3}}{27} \tag{18b}$$

where $\theta = 0$ and $\pi/3$ correspond to TC and TE, respectively.

Table 2. Comparison between various 3D versions of HB criterion

Strength criteria type	Reduction to HB criterion	Smoothness	Convexity	Explicit expression
Pan & Hudson (1988)	No	Yes	Yes	Yes
Simplified Priest (2005)	No	No	No	Yes
Comprehensive Priest (2005)	No	Yes	Yes	Yes
GZZ (Zhang & Zhu, 2007; Zhang, 2008)	Yes	No	No	Yes
Jiang *et al*. (2011) criterion 1	Yes	No	No	Yes
Jiang *et al*. (2011) criterion 2	Yes	No	No	Yes
Jiang *et al*. (2011) criterion 3	Yes	Yes	No	Yes
Jiang & Xie (2012)	Yes	Yes	No	Yes
Jiang & Zhao (2015)	Yes	No	Yes	Yes
Jiang (2017a)	No	Yes	Yes	Yes
Jiang (2017b)	No	No	Yes	Yes
Benz *et al*. (2008)	Yes	Yes	Yes	No
Lee *et al*. (2012)	Yes	Yes	Yes	No
Zhang *et al*. (2013)	Yes	Yes	Yes	No
Newly modified GZZ (this paper)	Yes	Yes	Yes	Yes

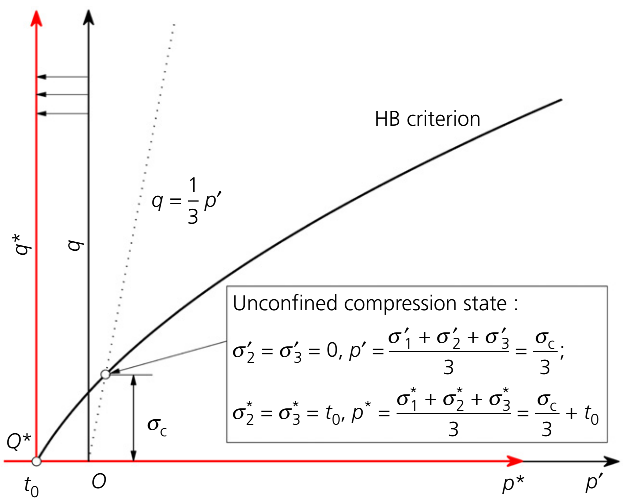

Fig. 2. Translation of stress space

Substituting equation (18) into equation (16) yields

$$f = \frac{1}{\sigma_{c,p}^{(1/a-1)}} \eta^{1/a} + \frac{m_b}{2}\eta - \frac{m_b}{2}\eta\sqrt{\frac{108 - 9\eta^2 - \eta^3\cos(3\theta)}{27\eta^2 - 9\eta^3\cos(3\theta)}} = 0 \tag{19}$$

where

$$\eta = \frac{q}{p^*} = \frac{9\tau_{oct}}{\sqrt{2}I_1^*} \tag{20a}$$

$$\sigma_{c,p} = \frac{\sigma_c}{p^*} = \frac{3\sigma_c}{I_1^*} \tag{20b}$$

The derivative of η with respect to θ is as follows:

$$\frac{\partial \eta}{\partial \theta} = -\frac{\partial f/\partial \theta}{\partial f/\partial \eta} = \frac{3m_b\eta^2(\eta^2 - 9)\sin(3\theta)}{((1/a)-1)(\eta/\sigma_{c,p})^{1/a-1}\chi\eta^2[\eta\cos(3\theta) - 3]^2 + m_b[54 - 27\eta\cos(3\theta) + \eta^3\cos(3\theta)]} \tag{21}$$

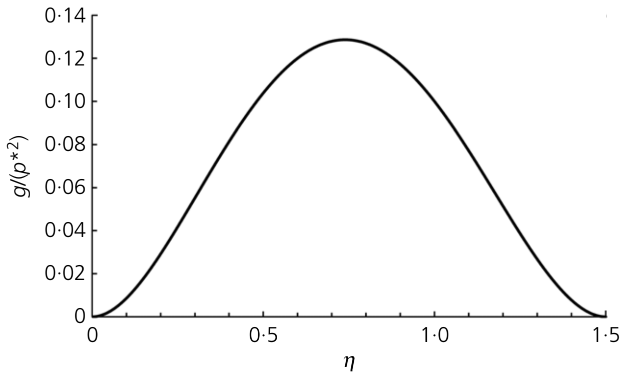

Fig. 3. Relation between g and η

where

$$\chi = \sqrt{\frac{108 - 9\eta^2 - \eta^3\cos(3\theta)}{3\eta^2 - \eta^3\cos(3\theta)}} \tag{22}$$

As noted earlier, the non-smoothness of the GZZ criterion only takes place at TE and TC. By checking equation (21), $\partial\eta/\partial\theta$ equals zero at $\theta = 0$ (TC) and $\pi/3$ (TE). Therefore, the proposed criterion is smooth under both TC and TE.

Equation (6) can be used to check the convexity of the proposed criterion. The criterion is convex at TC where it reaches the largest radius. Therefore, only the convexity at TE will be checked.

According to the chain rule, the second derivative of η at TE ($\theta = \pi/3$) is

$$\frac{\partial^2\eta}{\partial\theta^2}\left(\theta = \frac{\pi}{3}\right) = -\frac{9m_b(\eta^2 - 9)\eta^2}{((1/a)-1)(\eta/\sigma_{c,p})^{1/a-1}\eta^2(\eta+3)^2\chi(\pi/3) + m_b(54 + 27\eta - \eta^3)} \tag{23}$$

With $\rho = \eta p^*$ and equation (23), equation (6) can be rewritten as

$$g = \rho^2 + 2\left(\frac{\partial \rho}{\partial \theta}\right)^2 - \rho\frac{\partial^2 \rho}{\partial \theta^2} = (\eta p^*)^2\left[1 + \frac{9m_{\mathrm{b}}\left(\eta^2 - 9\right)\eta}{\left(\dfrac{1}{a} - 1\right)\left(\dfrac{\eta}{\sigma_{\mathrm{c,p}}}\right)^{1/a-1}\eta^2(\eta+3)^2\chi\left(\dfrac{\pi}{3}\right) + m_{\mathrm{b}}\left(54 + 27\eta - \eta^3\right)}\right] \tag{24}$$

To check the sign of equation (24), the range of η must be determined. Under TE, σ^*_{m} and equation (16) can be simplified as

$$\sigma^*_{\mathrm{m}} = \frac{\sigma^*_1 + \sigma^*_3}{2} = \frac{2\sigma^*_1 + \sigma^*_3}{3} - \frac{\sigma^*_1 - \sigma^*_3}{6} = p^* - \frac{q}{6} \tag{25a}$$

$$f = \frac{1}{\sigma_{\mathrm{c,p}}^{(1/a-1)}}\eta^{1/a} - m_{\mathrm{b}}\left(1 - \frac{2\eta}{3}\right) = 0 \tag{25b}$$

Equation (25b) can be further written as

$$1 - \frac{2\eta}{3} = \frac{1}{m_{\mathrm{b}}\sigma_{\mathrm{c,p}}^{(1/a-1)}}\eta^{1/a} > 0, \ \text{for}\ \eta > 0 \tag{26}$$

Hence, the upper boundary of η is 1·5 and thus $0 < \eta < 1{\cdot}5$. Considering this and $a \leq 1$, equation (24) can be rewritten as

$$g > (\eta p^*)^2\left[1 + \frac{9m_{\mathrm{b}}\left(\eta^2 - 9\right)\eta}{m_{\mathrm{b}}\left(54 + 27\eta - \eta^3\right)}\right] = (\eta p^*)^2\frac{8\eta^3 + 54 - 54\eta}{54 + 27\eta - \eta^3} \tag{27}$$

By plotting g against η based on equation (27) in Fig. 3, g is always larger than 0 when $0 < \eta < 1{\cdot}5$. Therefore, the proposed criterion satisfies equation (6) and is thus convex under TE.

Accuracy for prediction of rock strength

To check the accuracy of the newly modified GZZ criterion in predicting the strength of rock, the true triaxial test data of 11 intact rocks were collected, as summarised in Table 3 and

Table 3. Back calculated σ_{c}, m_{i}, a and s of 11 rocks

Rock	σ_{c}: MPa	m_{i}	s	a	References
KTB amphibolite	165	36·4	1	0·5	Chang & Haimson (2000)
Westerly granite	201	37·9	1	0·5	Haimson & Chang (2000)
Dunham dolomite	257	10·1	1	0·5	Mogi (1971)
Apache leap tuff	147	17·9	1	0·5	Wang & Kemeny (1995)
Mizuho trachyte	100	10·6	1	0·5	Mogi (1971)
Coconino sandstone	56·1	24·9	1	0·5	Ma & Haimson (2016)
Solnhofen limestone	310	4·96	1	0·5	Mogi (1971)
Yamaguchi marble	82	10·1	1	0·5	Mogi (1971)
Manazuru andesite	140	34·2	1	0·5	Mogi (2007)
Inada granite	229	29·8	1	0·5	Mogi (2007)
Orikabe monzonite	234	19·8	1	0·5	Mogi (2007)

Table 4. Accuracy evaluation of various 3D versions of HB criterion for strength prediction

Rock		Benz *et al.* (2008)	Lee *et al.* (2012)	Zhang *et al.* (2013) using H-D	Newly modified GZZ
KTB Amphibolite	NRMSD	*0·9861*	**0·9900**	0·9849	***0·9901***
	R^2	*0·0405*	**0·0341**	0·0422	***0·0340***
Westerly granite	NRMSD	*0·9969*	**0·9985**	0·9964	***0·9985***
	R^2	*0·0216*	**0·0149**	0·0233	***0·0148***
Dunham dolomite	NRMSD	**0·9822**	0·9661	***0·9840***	*0·9727*
	R^2	**0·0230**	0·0317	***0·0218***	*0·0285*
Apache leap tuff	NRMSD	**0·9727**	0·9677	***0·9734***	*0·9704*
	R^2	**0·0506**	0·0551	***0·0500***	*0·0527*
Mizuho trachyte	NRMSD	**0·9586**	***0·9600***	*0·9581*	0·9529
	R^2	**0·0365**	***0·0359***	*0·0368*	0·0389
Coconino sandstone	NRMSD	*0·9805*	***0·9849***	0·9785	**0·9833**
	R^2	*0·0485*	***0·0427***	0·0511	**0·0449**
Solnhofen limestone	NRMSD	*0·9358*	***0·9506***	0·9291	**0·9470**
	R^2	*0·0280*	***0·0246***	0·0295	**0·0255**
Yamaguchi marble	NRMSD	**0·9175**	0·8968	***0·9246***	*0·8973*
	R^2	**0·0892**	0·0997	***0·0852***	*0·0995*
Manazuru andesite	NRMSD	**0·9226**	***0·9309***	*0·9203*	0·9126
	R^2	**0·0832**	***0·0786***	*0·0845*	0·0884
Inada granite	NRMSD	*0·9943*	**0·9953**	0·9938	***0·9966***
	R^2	*0·0228*	**0·0206**	0·0238	***0·0177***
Orikabe monzonite	NRMSD	**0·9860**	0·9808	*0·9859*	***0·9883***
	R^2	**0·0315**	0·0368	*0·0317*	***0·0288***

Note: The ***bold italic*** numbers denote the highest accuracy prediction for a rock; the **bold** numbers denote the second highest accuracy prediction for a rock, and the *italic* numbers denote the third highest accuracy prediction for a rock.

detailed in the supplementary data. The newly modified GZZ criterion and other three 3D HB criteria (Benz *et al.*, 2008; Lee *et al.*, 2012; Zhang *et al.*, 2013), all of which can reduce to the HB criterion and are smooth and convex, are used to predict the strength of rock.

The rocks were tested in the laboratory using intact rock specimens, and thus $s = 1$, $a = 0{\cdot}5$ and $m_b = m_i$. To use a criterion to predict the strength, σ_c and m_i should be known. In this regard, the HB criterion was used to fit the triaxial test data to determine σ_c and m_i. The determined σ_c and m_i of the 11 rocks are shown in Table 3.

To quantitatively evaluate the accuracy of different criteria, the coefficient of determination (R^2) and normalised root-mean-square deviation (NRMSD) were calculated

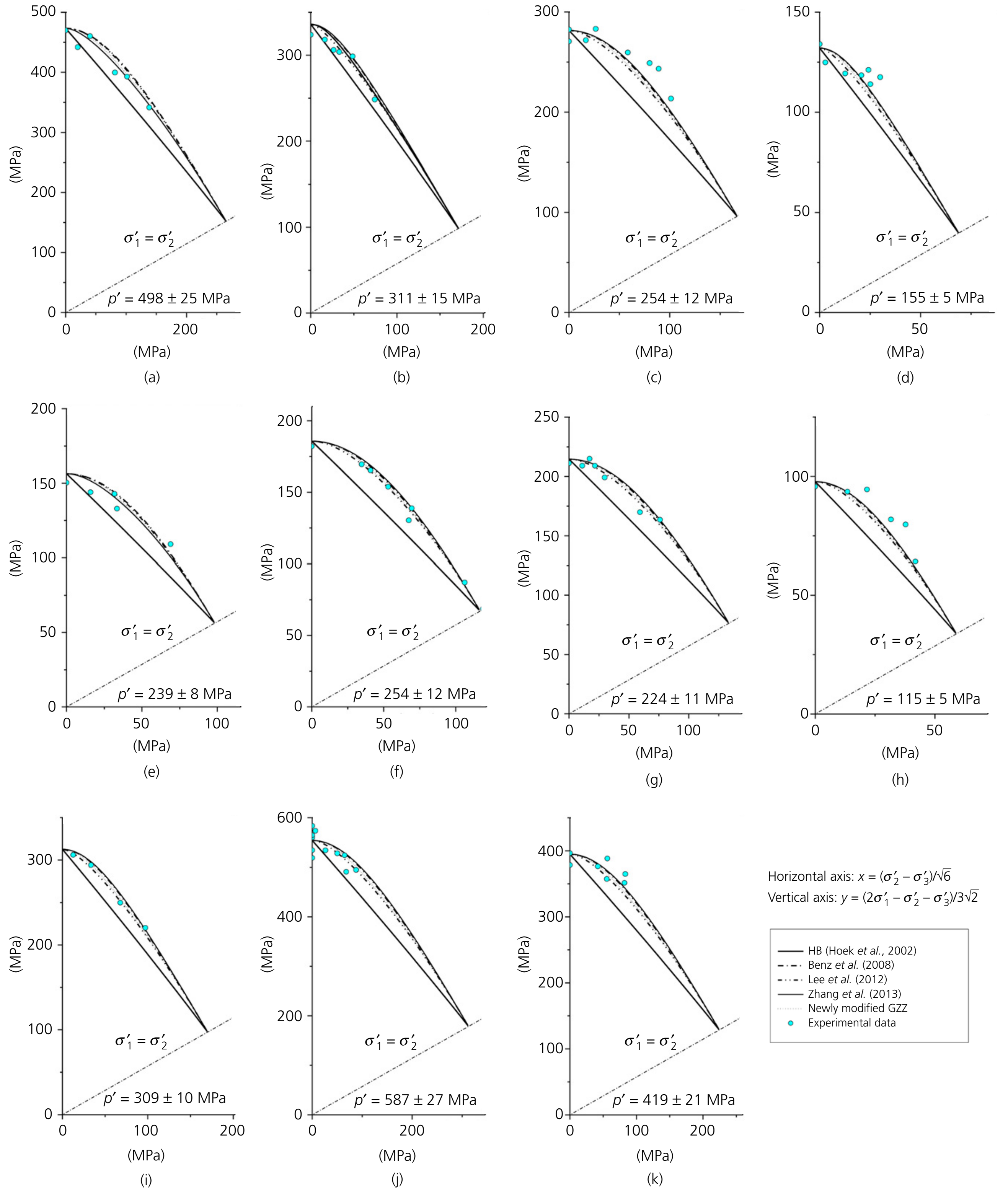

Fig. 4. Comparison of experimental data and prediction results by HB criterion (Hoek *et al*., 2002), Benz *et al*. (2008) criterion, Lee *et al*. (2012) criterion, Zhang *et al*. (2013) criterion and the newly modified GZZ criterion in π-plane: (a) KTB amphibolite; (b) Westerly granite; (c) Dunham dolomite; (d) Apache leap tuff; (e) Mizuho trachyte; (f) Coconino sandstone; (g) Solnhofen limestone; (h) Yamaguchi marble; (i) Manazuru andesite; (j) Inada granite; (k) Orikabe monzonite

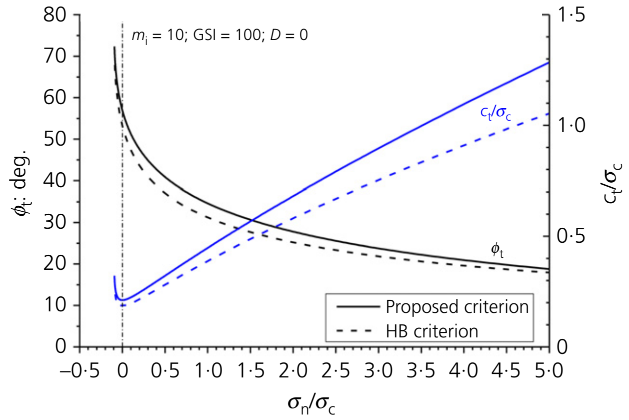

Fig. 5. Relationship among ϕ_t, c_t and σ_n' for proposed and HB criteria under a plane strain condition

by using the test and predicted τ_{oct} values. Table 4 shows the obtained R^2 and NRMSD values of the 11 rocks using the four different criteria. Only the $L(\theta_\sigma)_{\text{H-D}}$ was considered when using the Zhang *et al.* (2013) criterion to avoid repeating. The numbers of the highest, second-highest and third-highest accuracy predictions are 0, 6 and 5 for the Benz *et al.* (2008) criterion, 4, 3 and 0 for the Lee *et al.* (2012) criterion, 3, 0 and 3 for the Zhang *et al.* (2013) criterion and 4, 2 and 3 for the newly modified GZZ criterion, respectively. If one assigns a score of 3, 2 and 1 to the highest, second-highest and third-highest accuracy predictions, the total score will be 17, 18, 12 and 19 for the Benz *et al.* (2008) criterion, the Lee *et al.* (2012) criterion, the Zhang *et al.* (2013) criterion and the newly modified GZZ criterion, respectively. Therefore, it can be concluded that the newly modified GZZ criterion has the highest or at least equivalent accuracy among the various 3D HB criteria in predicting the strength of the 11 rocks.

The prediction accuracy of the four 3D HB criteria can be further seen from the π-plane plots in Fig. 4. Since the true triaxial tests were conducted under different p' values, the test data with p' values within 5% difference from the average are selected. The HB criterion underestimates the strength under true triaxial stress conditions because σ_2' is not considered. All the four 3D HB criteria, especially the newly modified GZZ criterion, however, predict the strength with a good accuracy.

Determination of tangential friction angle and cohesion for practical applications

Compared with other 3D HB criteria, the proposed criterion has an explicit expression and can be more easily applied to theoretical and numerical analyses. For the applications of the HB criterion in practice (Serrano & Olalla, 2004; Yang *et al.*, 2004), the non-linear relations between tangential friction angle ϕ_t, cohesion c_t and effective normal stress σ_n on the failure surface are major differences between the HB criterion and the MC criterion. The ϕ_t and c_t are generally defined as

$$\tan\phi_t = \frac{d\tau}{d\sigma_n} \tag{28a}$$

$$c_t = \tau - \sigma_n \tan\phi_t \tag{28b}$$

where τ is the shear stress on the failure surface.

Since there are no explicit expressions for the other 3D HB criteria (Benz *et al.*, 2008; Lee *et al.*, 2012; Zhang *et al.*, 2013), the ϕ_t based on equation (28a) cannot be easily determined from the equations shown in Table 1. In this regard, the proposed criterion is more suitable than the other 3D HB criteria for practical applications such as analyses of rock socketed piles (Serrano & Olalla, 2004) and rock slope stability (Yang *et al.*, 2004). To better understand the difference between the proposed criterion and the HB criterion, Fig. 5 shows the relations between ϕ_t, c_t and σ_n under a plane strain condition for the two criteria. As can be seen, the proposed criterion gives higher ϕ_t and c_t than the HB criterion due to the consideration of σ_2'.

CONCLUSIONS

This paper presents a simple modification of the GZZ criterion to address the non-smoothness and non-convexity problems. The major conclusions can be summarised as follows:

(*a*) Although various 3D HB criteria were proposed to consider the effect of σ_2', most of them cannot reduce to the HB criterion or are not smooth or convex at TE/TC. A few of the 3D HB criteria can reduce to the HB criterion and are smooth and convex, but they cannot be expressed explicitly.

(*b*) By introducing a 3D version of mean effective stress σ_m', the GZZ criterion was further modified. The newly modified GZZ criterion successfully addresses the non-smoothness and non-convexity problems while maintaining the expected features such as explicit expression.

(*c*) The newly modified GZZ criterion has the highest or at least equivalent accuracy in predicting the strength of 11 rocks among the 3D HB criteria that can reduce to the HB criterion and are smooth and convex. Furthermore, compared with the other 3D HB criteria, the newly modified GZZ criterion, due to its explicit form, can be more conveniently used to determine the tangential friction angle and cohesion on the failure surface for theoretical and numerical analyses.

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